

Math 421

Friday, December 5

Announcements

- Homework 10 due **today** (last HW!)
- Final Exam –
 - Date: Monday, December 15th
 - Time: 5:05-7:05 pm
 - Location: Bardeen 140
- McBurney & Makeups
 - McBurney exams – Sign up through the Testing Center (RegisterBlast) early!
 - Makeup exams – If you emailed me about taking a makeup final – you should have received the date & time. If you did not, email me again.

Upper & Lower Sums



Definition. Let $f: [a, b] \rightarrow \mathbb{R}$ be bounded and $P = \{t_0, t_1, \dots, t_{n-1}, t_n\}$ be a partition of $[a, b]$. The **upper and lower (Darboux) sums** for P are defined by

$$U(f, P) = \sum_{k=1}^n \left[M_k (t_k - t_{k-1}) \right] \quad M_k = \sup\{f(x) : x \in [t_{k-1}, t_k]\}$$

(supremum of the function of the subinterval) ·
(length of the subinterval)
often maximum

$$L(f, P) = \sum_{k=1}^n m_k (t_k - t_{k-1}) \quad m_k = \inf\{f(x) : x \in [t_{k-1}, t_k]\}$$

often minimum

$$P = \{-1, 2, 4\}$$

$$[-1, 2], [2, 4]$$

$$P = \{-1, 4\}$$

$$[-1, 4] \quad I_P$$

$$Q = \{-1, 0, 2, 4\}$$

$$[-1, 0], [0, 2], [2, 4] \quad I_Q$$

Relationship Between Partitions

Lemma 32.2. If P, Q are partitions of $[a, b]$ such that $P \subseteq Q$ and f is bounded on $[a, b]$, then

$$L(f, P) \leq L(f, Q) \leq U(f, Q) \leq U(f, P)$$

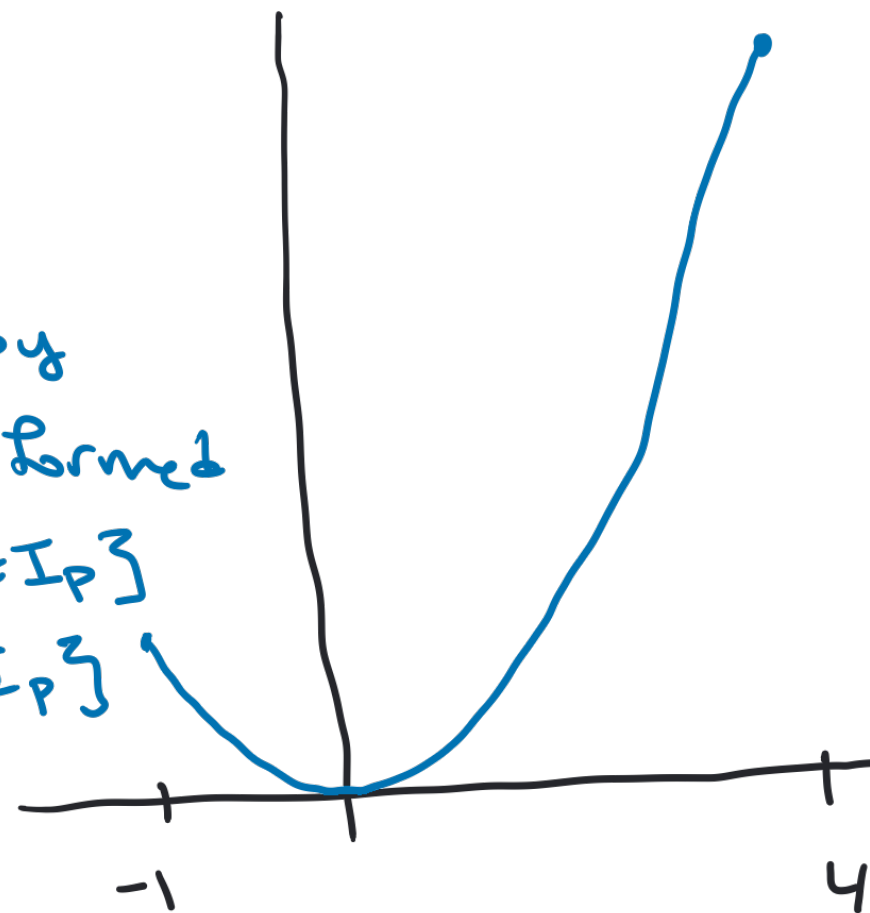
Discussion.

If $P \subseteq Q$ (partitions) then

$$f(x) = x^2$$

$$[-1, 4]$$

$I_Q \subseteq I_P$ (every sub interval formed by Q is a subset of some subinterval formed by P).
 $\inf \{f(x) : x \in I_Q\} \geq \inf \{f(x) : x \in I_P\}$
 $\sup \{f(x) : x \in I_Q\} \leq \sup \{f(x) : x \in I_P\}$



$$f(x) = x^2$$

$$[-1, 4]$$

$$P_1 = \{-1, .5, 4\}$$

$$P_2 = \{-1, 0, 3, 4\}$$

Lower Sums $\leq$ Upper Sums

$b_0 = a, t_n = b$
 $\rightarrow$ set $\{t_0, \dots, t_n\}$ $t_0 < t_1 < t_2 \dots < t_n$

Lemma 32.3. Let P_1 and P_2 be partitions of $[a, b]$ and let f be a function which is bounded on $[a, b]$. Then $L(f, P_1) \leq U(f, P_2)$.

Proof. Want to use lemma 32.2.

Define $Q = P_1 \cup P_2$ so $P_1 \subseteq Q, P_2 \subseteq Q$. By lemma 32.2

$$L(f, P_1) \leq L(f, Q) \leq U(f, Q) \leq U(f, P_2)$$

Thus $L(f, P_1) \leq U(f, P_2)$

Upper & Lower Integrals

Definition. Let $f: [a, b] \rightarrow \mathbb{R}$ be bounded. The **upper (Darboux) integral** and the **lower (Darboux) integral** of f over $[a, b]$ are defined by

$$\rightarrow \sum_{i=1}^n M_i \cdot (t_i - t_{i-1})$$

- $U(f) = \inf\{U(f, P) : P \text{ is a partition of } [a, b]\}$

- $L(f) = \sup\{L(f, P) : P \text{ is a partition of } [a, b]\}$

will be finite
 $\rightarrow \{a, b\}$

• $f(x)$ is bounded on $[a, b]$

$$P \subseteq Q$$

$$L(f, P) \leq L(f, Q) \leq U(f, Q) \leq U(f, P)$$

$$\swarrow$$

$$\inf \{f(x) : x \in [a, b]\} \cdot (b-a)$$

$$\nwarrow$$

$$\sup \{f(x) : x \in [a, b]\} \cdot (b-a)$$

$$L(f, P_1) \leq U(f, P_2)$$

$$L(f) \leq U(f)$$

$$L(f) = \sup \{ L(f, P) : P \text{ is a partition of } [a, b] \}$$

$$U(f) = \inf \{ U(f, P) : " \}$$

Theorem 32.4. If f is a bounded function on $[a, b]$, then $L(f) \leq U(f)$. Fix a partition P of $[a, b]$

Proof. From lemma 32.3, $L(f, P)$ is a lower bound

for every upper sum so $L(f, P) \leq U(f)$. Thus

$U(f)$ is an upper bound for every lower sum, so

$$U(f) \geq L(f).$$

Integrable

• Theorem 32.4 says $L(f) \leq U(f)$
often just prove $U(f) \leq L(f)$.

Definition. A function f which is bounded on $[a, b]$ is **integrable** on $[a, b]$ if

$$L(f) = U(f)$$

In this case, the value is called the **integral** of f and is denoted by

$$\int_a^b f = \int_a^b f(x) dx.$$

$$[a, b], \quad c > 0$$

Activity

For each of the following functions, the upper and lower integrals and determine if the function is integrable.

1. Define $f(x) = c$.

$$U(f) = (b-a)c$$

$$L(f) = (b-a)c$$

2. Define $g(x) = \begin{cases} c, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$

$$U(g) = (b-a)c$$

$$L(g) = 0$$

